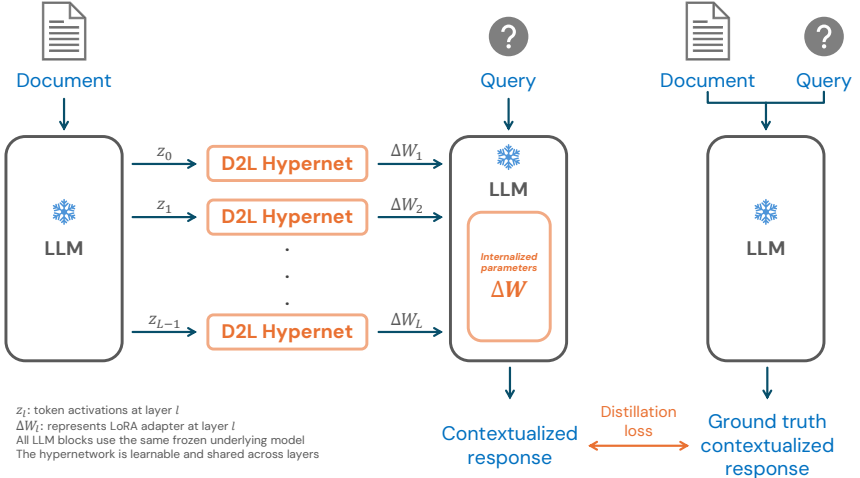
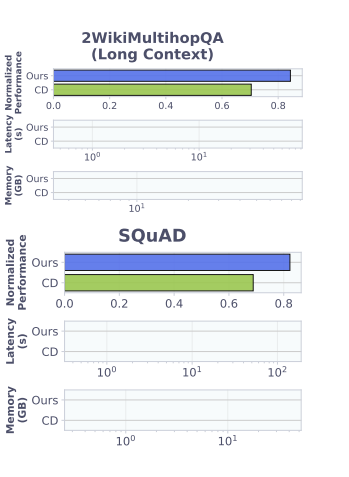

Document
Query
LLM
z_0
z_1
z_{L-1}
D2L Hypernet
D2L Hypernet
D2L Hypernet
ΔW_1
ΔW_2
ΔW_L
LLM
Internalized parameters
ΔW
z_l: token activations at layer l
ΔW_l: represents LoRA adapter at layer l
All LLM blocks use the same frozen underlying model
The hypernetwork is learnable and shared across layers
Contextualized response
Distillation loss
Document
Query
LLM
Ground truth contextualized response


2WikiMultihopQA (Long Context)
Normalized Performance
Ours
CD
0.0
0.2
0.4
0.6
0.8
Latency (s)
Ours
CD
10^0
10^1
Memory (GB)
Ours
CD
10^1
SQuAD
Normalized Performance
Ours
CD
0.0
0.2
0.4
0.6
0.8
Latency (s)
Ours
CD
10^0
10^1
10^2
Memory (GB)
Ours
CD
10^0
10^1